Snowflake

Real-Time Fraud Detection — Business Case

Prepared for Zilch • Head of Fraud • Executive Leadership

Confidential
Customer Engineering
July 2026

THE PROBLEM ZILCH IS TRYING TO SOLVE

Fraudsters test stolen cards in rapid bursts — often 5–10 low-value transactions within seconds. If your fraud model only sees **stale features**, every transaction in a burst looks clean until it's too late. The loss isn't just the fraudulent spend: it's chargebacks, issuer fees, reputational risk with Thredd, and the engineering cost of building the detection capability from scratch.

WHAT SNOWFLAKE DELIVERS — PROVEN IN TECHNICAL VALIDATION

<2s
Feature freshness out of the box

~14ms
End-to-end scoring per transaction

1
Platform replacing 7+ managed services

Zero
Custom ETL pipelines built or maintained

CURRENT COMPLEXITY VS. SNOWFLAKE APPROACH

WHAT ZILCH IS BUILDING

- ⚠ Feature freshness still in development — detection window is undefined until it ships
- ⚠ 7+ services to build, deploy & monitor: Kinesis, Lambda, Glue, RAPIDS, SageMaker, Triton, EKS
- ⚠ Custom preprocessing pipeline needed to align training and serving — drift risk on every retrain
- ⚠ Multiple security boundaries, credentials, and vendor relationships to manage
- ⚠ Engineering bandwidth consumed by infrastructure, not fraud strategy

WHAT YOU GET WITH SNOWFLAKE

- ✓ Sub-2s freshness with no custom build — declare features once, they stream continuously
- ✓ Single platform covers ingestion, feature store, model training, serving, and data warehouse
- ✓ Identical feature logic in training and serving — model drift is structurally impossible
- ✓ One security boundary, one billing relationship, one governance model
- ✓ Fraud team focuses on model quality and thresholds — not infrastructure uptime

PLATFORM SIMPLIFICATION

CURRENT AWS/NVIDIA STACK — 7+ SERVICES

Amazon Kinesis • AWS Lambda • Amazon S3 • AWS Glue ETL • RAPIDS Preprocessing • Amazon SageMaker • NVIDIA Triton • fraud-engine-service • Amazon EKS • Amazon DynamoDB

→

SNOWFLAKE — ONE PLATFORM

Snowpipe Streaming • Online Feature Store • Snowpark ML Training • Model Registry • SPCS Scoring Service

BUSINESS IMPACT — ESTIMATED COST SAVINGS

Faster Detection = Less Fraud Loss

Stale features show **zero velocity signal** during a card-testing burst. Real-time freshness detects the pattern from txn 2 onwards — blocking the follow-on £500 fraud event before it completes.

£660K–£780K / year
→ £1.3M–£1.6M at 2× volume
24M txns × 0.05% fraud rate × 20% card-testing share = £1.2M exposure. +55pp burst detection improvement vs. stale features (£500 avg incident cost).

⚙ Engineering Cost Reduction

The 10-service stack requires ~1.75 FTE-equivalents of ongoing maintenance. Snowflake reduces that to ~0.5 FTE. AWS infrastructure saving is modest at current volume — the primary driver is headcount.

£170K–£220K / year
→ £220K–£300K at 3× volume (infra grows, Snowflake scales gradually)
~1.25 FTE freed at £130K fully loaded (primary driver). AWS infra saving £7K–£14K/yr at current scale — validated from unit pricing across Kinesis, Glue, EKS, SageMaker, DynamoDB.

⚡ Faster Model Iteration

Monthly cycles replace quarterly — 8 additional iterations/year. Each 4-week faster response to a new attack pattern saves ~£25K. Each 0.1% false decline reduction recaptures ~£2M in legitimate GMV.

£200K–£300K / year
→ £400K–£600K at 2× volume
8 extra cycles/yr at avg 0.3% recall gain (£30K per 0.5pp at 12K annual incidents) + £25K per new attack pattern caught 4 wks earlier (3–4 patterns/yr). GMV upside from false decline reduction excluded.

WHY ACT NOW

Zilch's feature freshness component is still being built. **Every week it remains under construction is a week the fraud model runs blind to intra-session patterns.** Snowflake's Online Feature Store is in public preview today — the technical validation in this document was run on live infrastructure. Zilch can have sub-2-second freshness without waiting for a custom build to ship, and without the ongoing burden of maintaining it.

SUGGESTED NEXT STEPS

1. 30-Min Architecture Review
Map Zilch's Thredd event flow into the Snowflake ingestion path. Confirm data residency and PrivateLink routing.

2. POC on Zilch's Live Data
Run the validated pipeline on a sample of real transaction history. Measure freshness improvement against current baseline.

3. Commercial Sizing
Snowflake scales on consumption — no upfront spend. We provide a TCO comparison against the current AWS/NVIDIA stack.